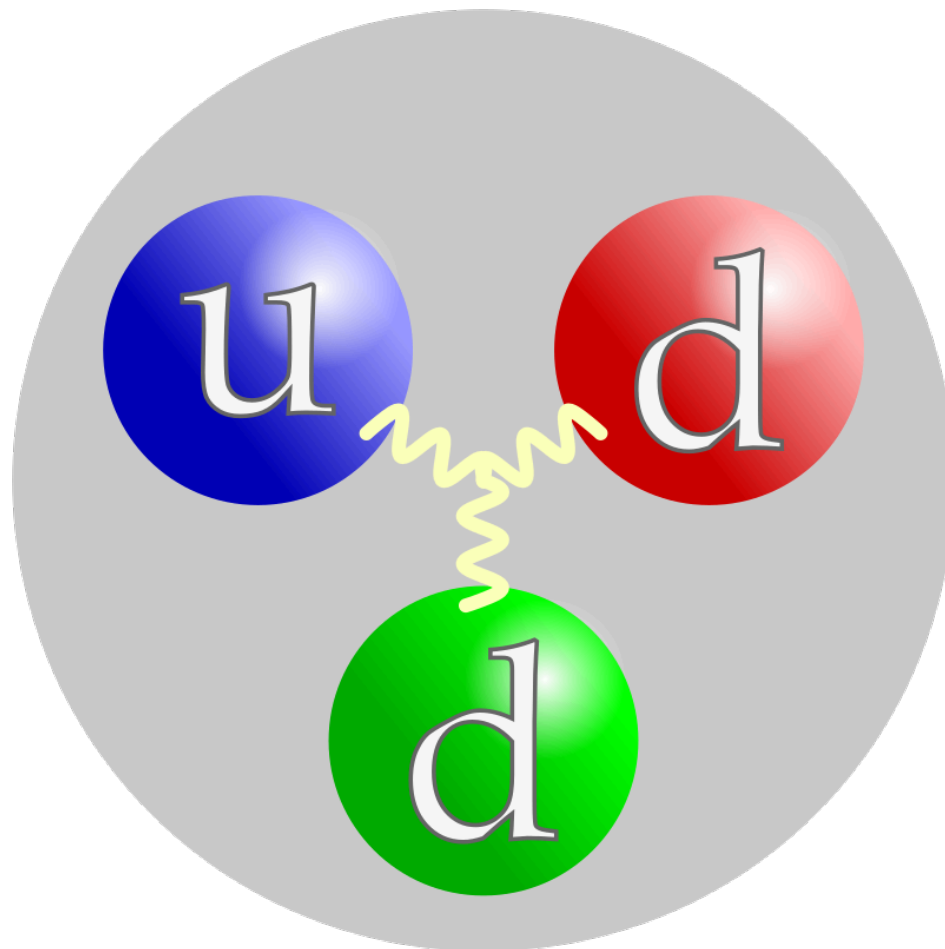


MAŁOKĄTOWE ROZPRASZANIE NEUTRONÓW



X-RAY VS. NEUTRON

	X-ray	Neutron
rozpraszane na	chmurze elektronowej	jądram atomowych
$SLD \propto Z$	tak	nie
rozpraszanie na H	w zasadzie 0	duże
rozróżnia H od D	nie	tak
znakowanie deuterem	nie	tak
czułe na $H_2O : D_2O$	nie	tak



SLD — Scattering Length Density

kontrast

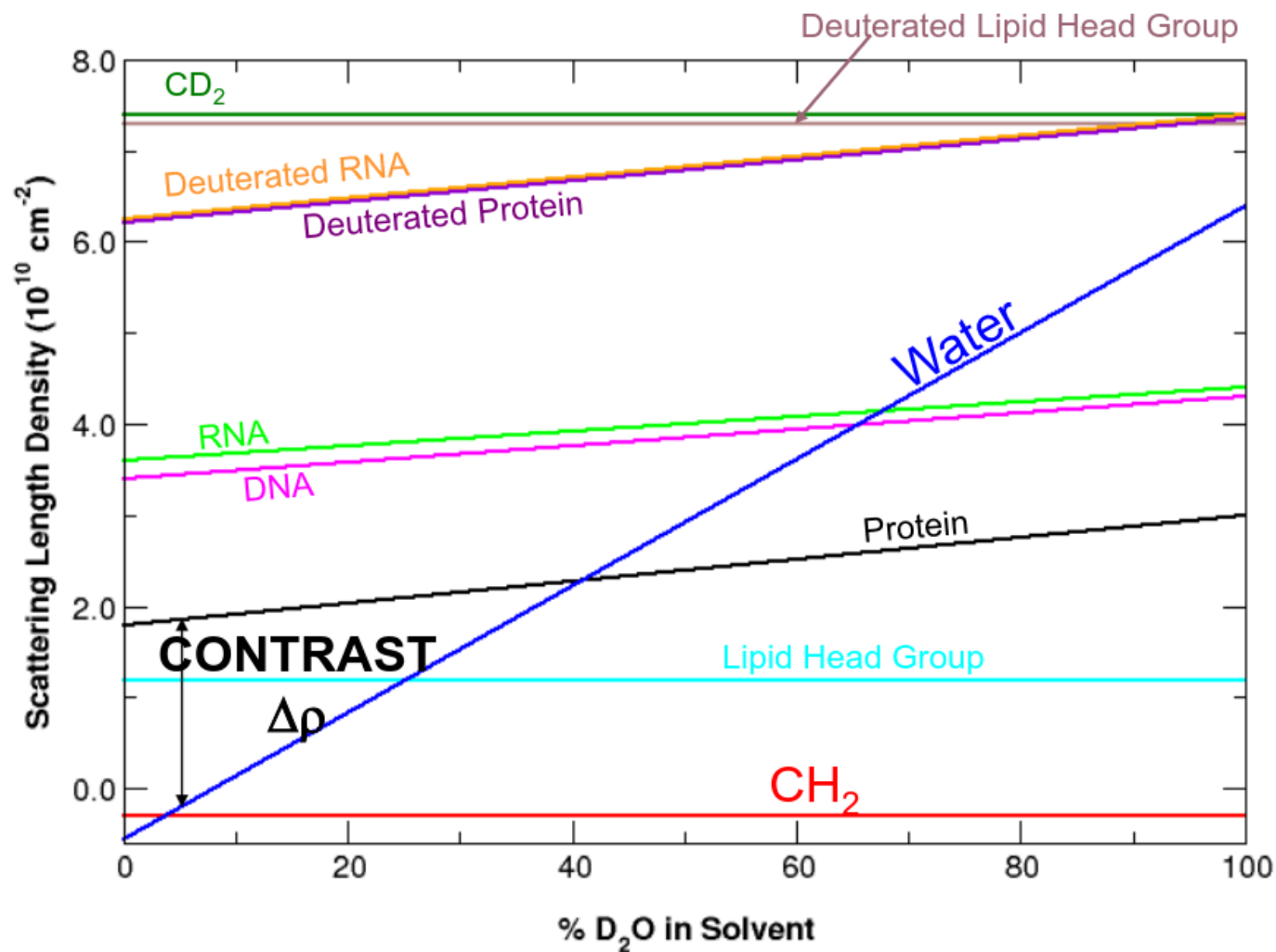
$$\rho = \frac{\sum b_i}{V_m}$$

$$\Delta\rho = \rho_{\text{molekuły}} - \rho_{\text{tła}}$$

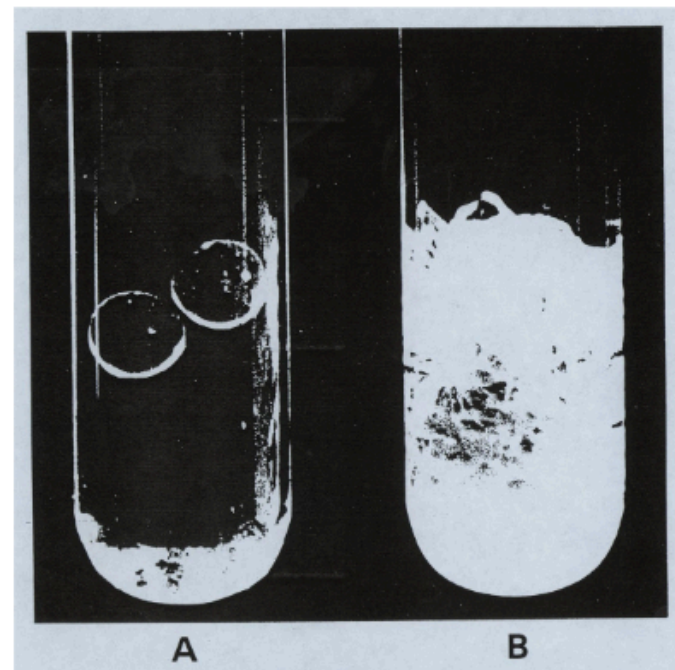
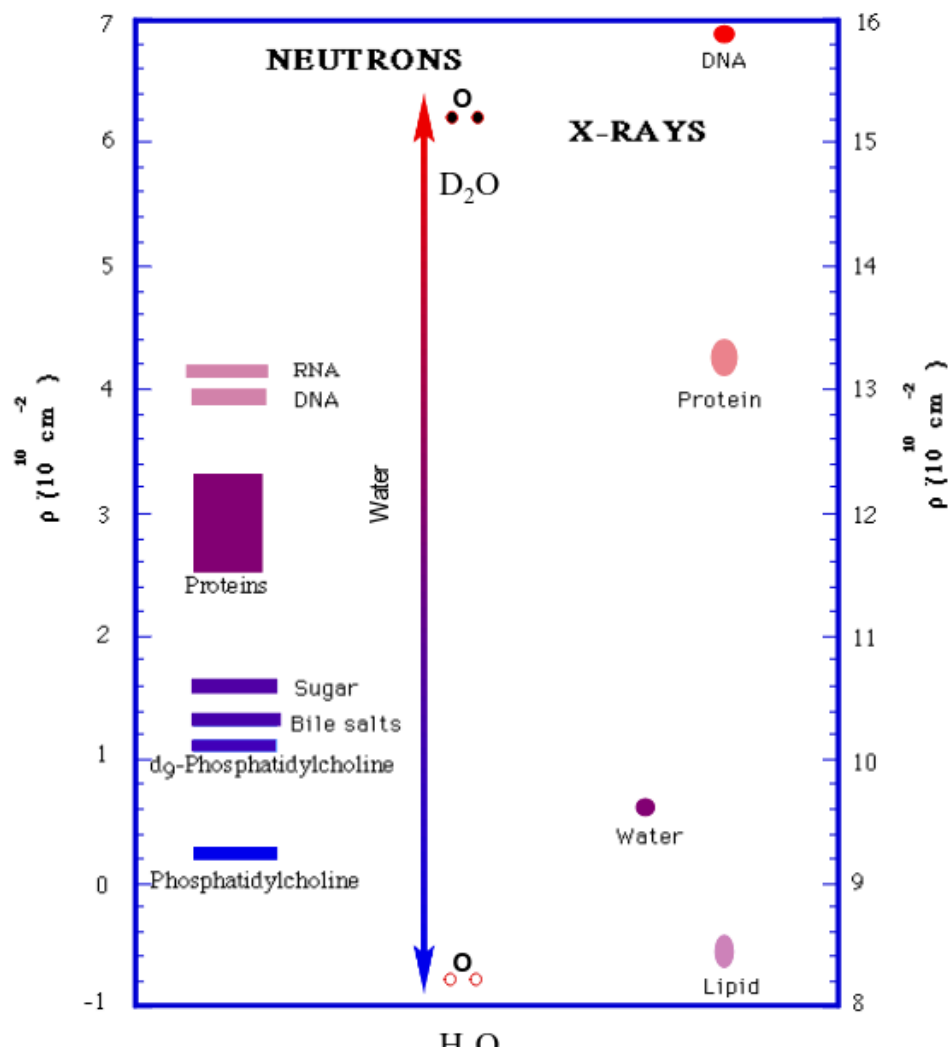
Contrast Variation $\Delta\rho \approx 0$ 

$$\Delta n = n_{\text{szkła}} - n_{\text{oleju}} \approx 0$$

SANS CONTRAST VARIATION, CONT.

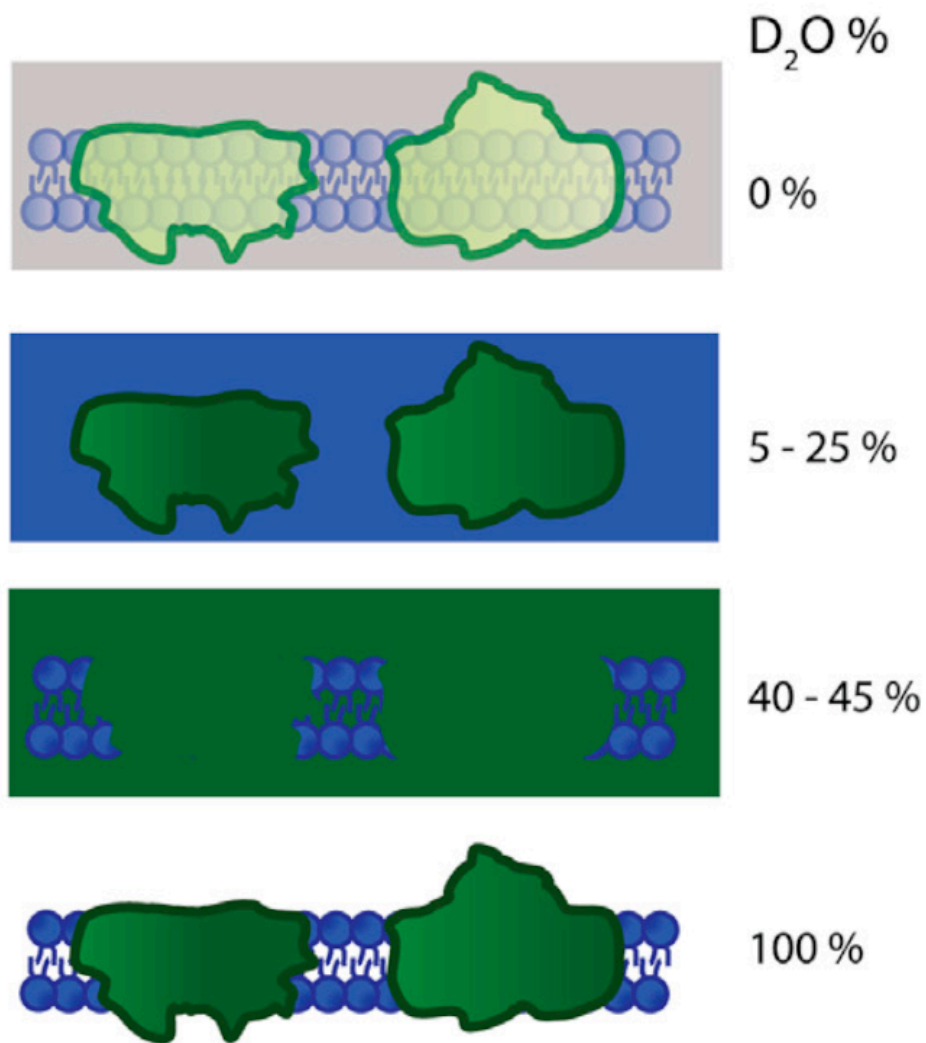


SANS CONTRAST VARIATION, CONT.

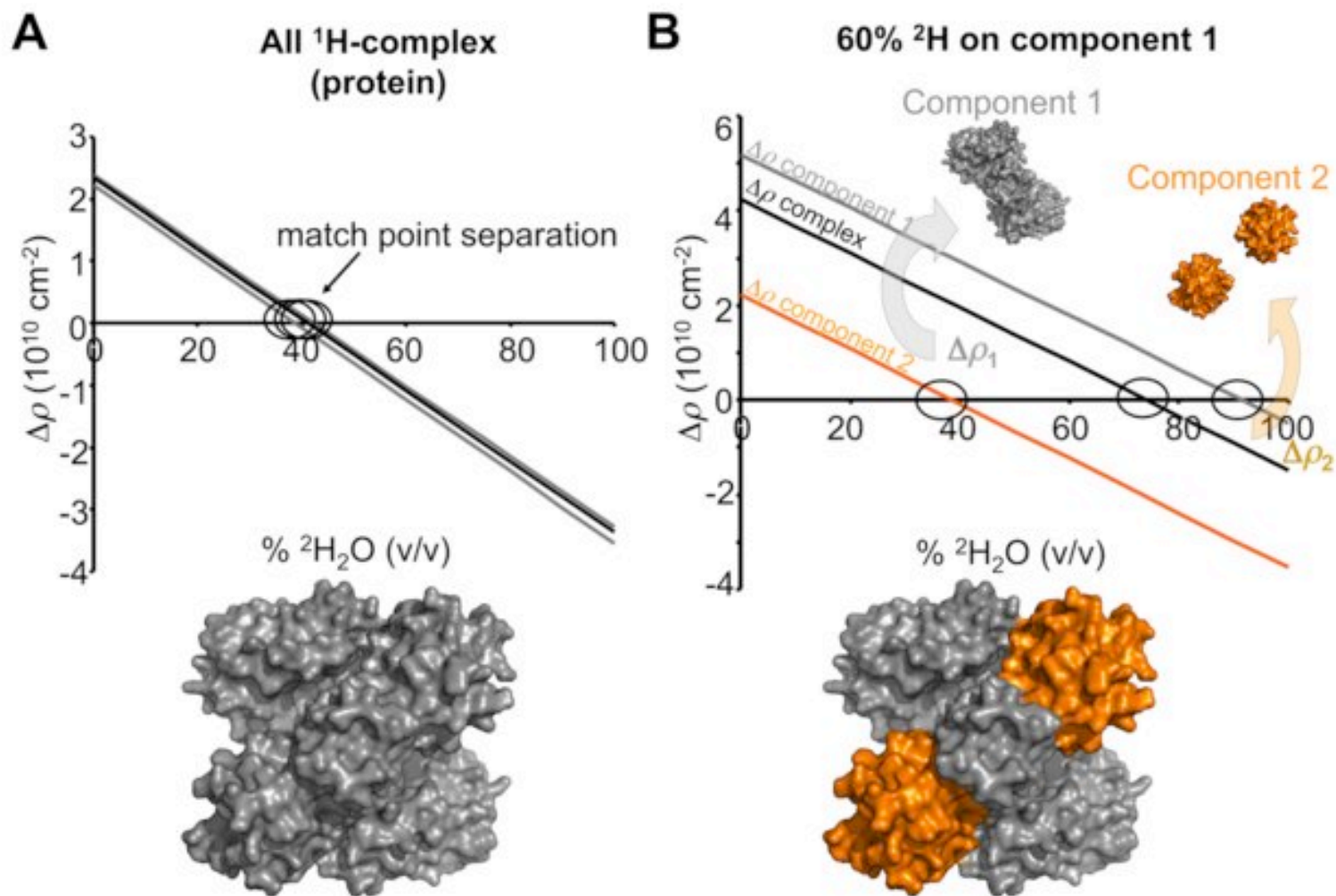


Both tubes contain borosilicate beads + pyrex fibers + solvent. (A) solvent refractive index matched to pyrex;. (B) solvent index different from both beads and fibers – scattering from fibers dominates

SELEKCJA SKŁADNIKÓW SYSTEMU



ZNAKOWANIE DEUTEREM



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